
Proposed CADIO-GWO Algorithm

Require: Objective function $f(x)$, population size N , maximum iterations T

Ensure: Best solution X_{best}

- 1: Initialize the population using Logistic chaotic mapping
 - 2: Evaluate the fitness of all search agents
 - 3: Determine the global best solution X_{best}
 - 4: **for** $t = 1$ to T **do**
 - 5: Compute the exponential adaptive parameter a
 - 6: Determine the adaptive elite population
 - 7: Sort the population according to fitness
 - 8: **for** each search agent **do**
 - 9: Randomly select an elite leader
 - 10: Update DIO position with adaptive global-best guidance
 - 11: Compute the elite-centroid guided pack position
 - 12: Generate the Grey Wolf guidance position
 - 13: Compute the adaptive hybridization weight
 - 14: Update the search agent using the final hybrid position
 - 15: Apply boundary constraints
 - 16: **end for**
 - 17: Evaluate the updated population
 - 18: Update X_{best}
 - 19: Store the best fitness value
 - 20: **end for**
 - 21: **return** X_{best}
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